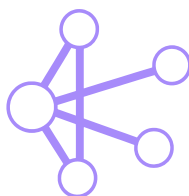


Portfolio Manager

A Long-Horizon Strategy for Crypto

Built on Ethereum Smart Contracts

September 22, 2026



Harris Partners Technologies

Campbell Harris

casper@harrispartners.tech · pm.harrispartners.tech

Abstract. HPT Portfolio Manager is a USDC-denominated vault giving investors exposure to an automated crypto trading strategy without requiring trust in the operator's bookkeeping. The *Trading System* is a private off-chain service: it maintains a curated, screened universe of liquid ERC-20 tokens, forecasts each with a pretrained time-series model (FinCast), and uses the *uncertainty* of those forecasts to select which exponential-moving-average lookback window governs a trend signal, gated and ranked by short-horizon momentum. The *Smart Contract Logic* is a set of Ethereum contracts that custody the assets, price them, mint and burn shares against a contract-computed NAV, and bound what the Trading System may do with investor capital. The split is deliberate: the strategy stays confidential, while every consequence of it (each trade, valuation, and fee) is published to a ledger anyone can audit without our permission. The value of the token is purely driven by the performance of the trading strategy, unlike other speculative token assets driven by the open market popularity contest.

Notice. This document describes a software system and its strategy. It is not an offer or solicitation for any security or financial product in any jurisdiction, nor investment, legal or tax advice. Past performance, including the backtests in \$4, does not predict future results; an investor may lose the entire deposit.

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1 Introduction

A systematic fund asks an investor for two distinct acts of trust. The first is trust in the strategy, that the model is sound. That one is unavoidable, and it is what the investor is paid to judge. The second is trust in the *reporting*: that the stated NAV is real, that trades happened at the prices claimed, that fees are honest, and that the money is still there. Traditional funds discharge the second through intermediaries such as administrators, auditors and custodians, who are themselves trusted parties, paid by the manager, reporting on a lag.

HPT's position is that the second act of trust is unnecessary. A strategy executed entirely by software, against natively digital assets, on a market that settles on a public ledger, can simply *publish its consequences* as they occur. Nothing needs to be attested to, because nothing needs to be reported. The record is the ledger.

The system splits along that line. Everything constituting proprietary insight lives off-chain and stays private; everything constituting a claim about investor money lives on-chain and is enforced by contract code. The **Trading System** decides what to hold. The **Smart Contract Logic** decides what the Trading System may do and what everything is worth. The Trading System never holds investor funds: it holds a key that can instruct the vault to trade through pre-approved routes, and nothing more.

2 Architecture

HPT is two pieces of software with a hard line drawn between them.

The **Trading System** is our own software, running privately and off the chain. It reads market data, runs the strategy on a fixed schedule, and decides what the fund should hold. It is the only component with a view of the model. It caches its own positions, but the authoritative record of holdings is always the vault's on-chain balances, against which it reconciles.

The **Smart Contract Logic** is the Ethereum code that holds the money. It takes custody of the assets, prices them, and issues and redeems shares. It knows nothing about the strategy and has no view on whether it is any good. It enforces a narrower property: that the Trading System can only move value through manager-allowlisted venues, that issuance and redemption follow a NAV computed on-chain, and that investor claims are honoured by code rather than discretion.

There are exactly three things the Trading System is allowed to do to the contracts, and each one needs a key the contracts recognise. It can *stage prices*, writing what each token is currently worth in USDC. It can *rebalance*, telling the vault to run a batch of trades against approved venues. And it can *settle*, working through the deposits and withdrawals investors have queued up with the vault directly. None of the three hands over custody. The Trading System cannot send a single token to an address of its own choosing; all it can do is ask the vault to act in ways that were approved beforehand.

Each cycle the Trading System syncs the tracked-token set, stages prices, reconciles its book, evaluates the strategy on closed daily bars, plans the difference between current and target holdings, and executes it as sliced orders. A faster loop runs alongside it to service the withdrawal queue, selling what it needs to before paying anyone out. Signals come only from *closed* bars, which is why live trading behaves exactly as the backtest did. The intraday price is used to value positions and size orders, but it never creates a signal.

3 The tradable universe

Trend-following is indiscriminate by nature: it will buy anything that has gone up. The universe definition is therefore not a preliminary to the strategy but one of its two risk controls, the other being position sizing.

The base universe is a hand-maintained watchlist naming each token, its ERC-20 address, and whether it is currently eligible. Inclusion is a human judgement reflecting what a screen cannot see: whether the asset is a credible ongoing concern, whether its contract does anything unusual, and whether its liquidity is real or merely incentivised. The eligibility flag is deliberately asymmetric. Marking an asset inactive does not merely stop new purchases, it forces an exit from any existing position on the next cycle, giving a manual override that needs no code change and no signal.

Because manual curation does not scale, a screener runs independently over Ethereum DEX pools and proposes candidates, filtering on tradability rather than price behaviour: minimum pool liquidity and 24-hour volume, so a position can be exited as well as entered; a bounded volume-to-liquidity ratio, which screens out implausible turnover relative to depth; minimum pool age, token history, and market capitalisation; and restriction to major quote assets. Survivors are scored and ranked, but the screener proposes rather than admits. Its output is a queue for human review, which preserves the curated core as the actual gate.

Passing that gate makes an asset *eligible*, not *held*. The book holds a fixed maximum number of positions and there are usually more eligible candidates than slots, so momentum enters the system twice doing two different jobs: as a *threshold* it is a veto, and as a *rank* it allocates the limited capital.

4 The trading strategy

4.1 The idea

The strategy follows trends on daily bars. Its core signal compares a fast exponential moving average of price against a slow one: fast above slow means the asset is in an uptrend and is worth holding. The difficulty is choosing the slow lookback. A short window reacts quickly but is whipsawed; a long one is stable but concedes much of each move before reacting. The usual responses are to fix the window by backtest and accept it is wrong whenever the regime changes, or to switch on realised volatility, which only describes the regime already survived.

HPT lets a forward-looking forecast make the choice. FinCast, a pretrained financial time-series model, is given roughly four months of daily closes and returns a probabilistic forecast over the following week. Two scalars are extracted: μ , the expected return, and σ , the forecast *uncertainty*, taken as the width of the predictive interval.

σ is not realised volatility, which rises only after turbulence has occurred; σ describes the model's confidence in what comes next and can widen *before* the move that justifies it. It is smoothed through the same two windows used for price: when forecast uncertainty is elevated relative to its own longer-run level the slower crossover governs, otherwise the faster one does. The switch is evaluated per asset per day.

What is *not* happening matters as much: the price forecast is never traded directly and no position is ever sized by μ . A foundation model's point forecast on a single crypto asset is not reliable enough to bet on; its uncertainty estimate, used to choose between two well-understood classical signals, is a far weaker claim to make of it, and a far more defensible one. If inference fails, the strategy falls back to rolling realised statistics and keeps trading.

4.2 Evidence

Figure 1 and Table 1 compare the adaptive rule against both fixed alternatives over the same universe and period, a ten-slot portfolio from January 2023 to January 2026, under low and high per-trade costs.

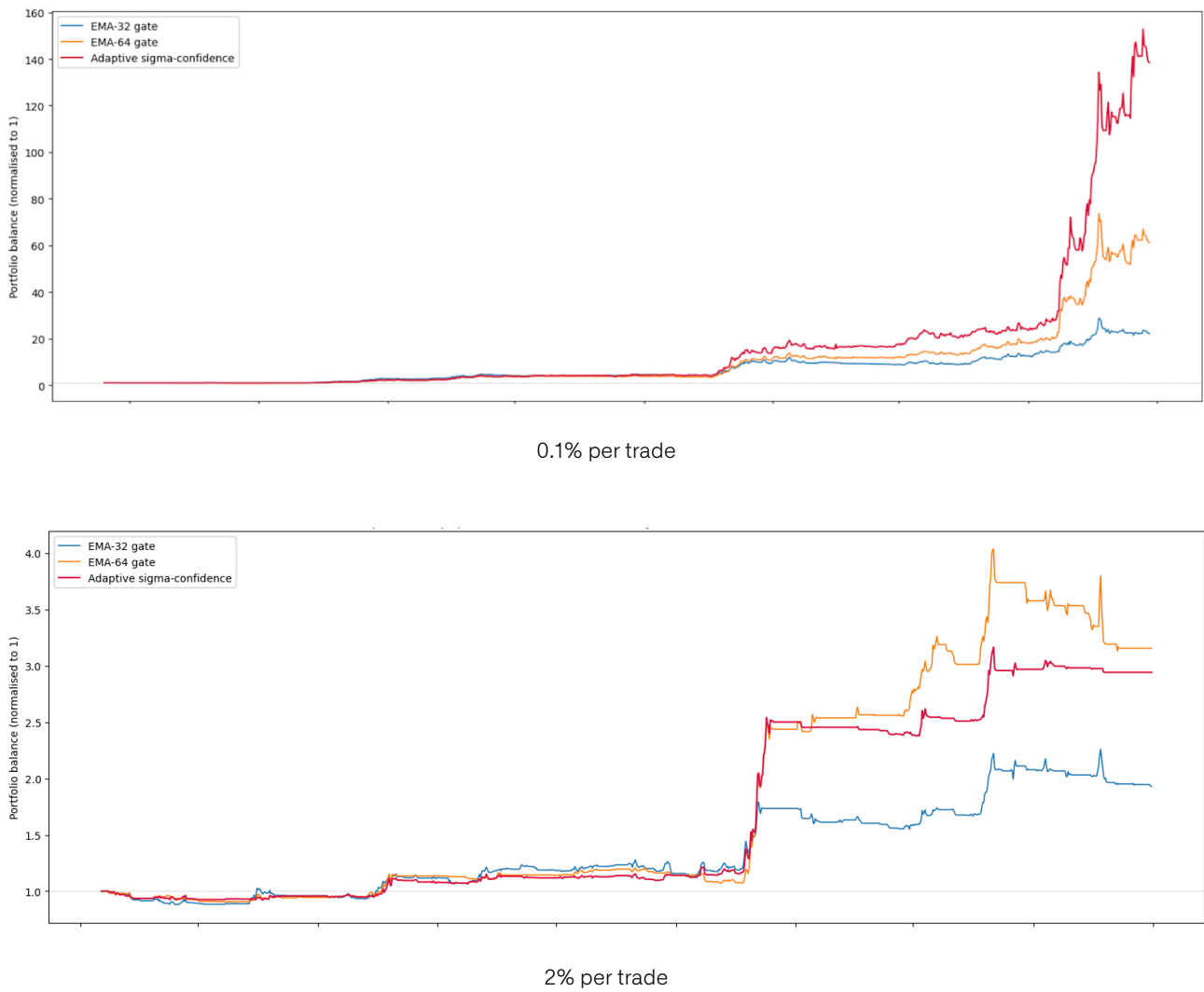


Figure 1. Equity curves, normalised to 1, Jan 2023 to Jan 2026. Note the differing vertical scales: the upper panel ends near 140x, the lower near 3x. Return is acutely sensitive to execution cost.

Lookback rule	0.1% per trade		2% per trade	
	Sharpe	Max DD	Sharpe	Max DD
Fixed short-slow (EMA-32)	2.629	-27.7%	1.131	-14.6%
Fixed long-slow (EMA-64)	2.872	-29.7%	1.629	-22.4%
Adaptive (FinCast σ)	3.067	-19.9%	1.701	-8.3%

Table 1. Risk-adjusted performance under both cost assumptions.

The adaptive rule has the best Sharpe ratio and the shallowest drawdown in both cost regimes, and its clearest contribution is to drawdown rather than to return: at 2% cost it cuts peak-to-trough loss to 8.3% against 22.4% for the fixed long window. That result should be read precisely, because the same panel shows the fixed long window finishing at a slightly *higher* terminal multiple, roughly 3.15 $\times$ against 2.95 $\times$. The honest summary is that letting forecast uncertainty select the lookback buys materially better risk-adjusted performance and a much smoother path, not uniformly higher raw return.

4.3 Entries, exits, and sizing

Momentum, measured as trailing return over a short window, is layered on top. A position is opened only when the trend signal has just *turned* positive (off on the prior bar, on now, so the strategy buys transitions rather than uptrends it has already missed), *and* μ is positive, *and* trailing momentum exceeds a positive threshold. It is closed when the signal turns off, immediately and with no grace period, or when momentum falls below a negative threshold and a minimum holding period has elapsed. Trend deterioration is acted on without delay, while the momentum exit is buffered so one bad day cannot round-trip a fresh position. When more assets qualify than there are slots, candidates are ranked by momentum and the strongest fill them.

The book is a fixed number of equally-weighted slots, each holding an asset or cash, filling and vacating independently, so the strategy never concentrates and unfilled slots leave it self-deleveraging in downtrends. Positions that drift from target weight are trimmed or topped up rather than closed and reopened. A minimum history requirement, a post-sale cooldown, a price floor, and the manual flags of \$3 all gate purchases.

5 Execution

Figure 1 makes the case for this section: the same signals return roughly 140x at 0.1% per trade and under 3x at 2%. Execution quality is not a detail of the strategy, it is most of it.

Orders are therefore not sent as single trades. Each buy or sell is sliced into chunks of randomised size with randomised delays: time-weighted execution that lets the pool refill between chunks, with randomisation making the sequence harder to anticipate. Chunk size is a genuine optimisation rather than a constant, because slicing finer reduces impact but multiplies fixed gas cost, so there is an interior optimum, estimated from measured impact against the actual pool. It is one number per token, so a larger position gets *more* chunks rather than larger ones.

Before trading, the Trading System compares an independent reference price against the on-chain quote, rejecting trades when impact exceeds a ceiling, slippage exceeds a bound, gas would consume too much of the trade, or the size is uneconomic. The prices the vault uses to value holdings are pushed on-chain by the Trading System and derived from what a position could actually be *liquidated* for, not from a headline mid-price, since valuing a thin position at mid overstates NAV and silently transfers value from redeeming holders to remaining ones. Prices are staged in the same transaction flow as the operation consuming them, so a valuation used for settlement is never older than the settlement. Relatedly, when the withdrawal queue needs more USDC than the vault holds, the Trading System sells to cover *before* settling: investors are never paid out of a valuation, only out of a realised sale.

6 The Smart Contract Logic: custody and constraints

The vault is custodian and accountant. NAV is computed on-chain on demand as the USDC balance plus the oracle-priced value of every tracked token held, excluding USDC escrowed against unsettled deposits and USDC reserved against settled but unclaimed withdrawals. Neither belongs to continuing shareholders, and including them would let queued activity distort the share price, which follows as NAV over shares outstanding.

Subscriptions and redemptions are two-phase. An investor calls the vault directly to request a deposit, USDC is escrowed immediately, and shares are minted later when the Trading System settles the queue, at a price the contract computes at that moment. A withdrawal request escrows shares; settlement

snapshots the USDC owed and fee due; the investor claims afterwards. Requests are cancellable before settlement. The design exists because valuation and liquidity are not instantaneous for a traded portfolio: it is a dealing cycle whose price is computed by code rather than struck by an administrator.

Execution is constrained. The Trading System submits a batch of calls, which the vault executes only against manager-allowlisted targets, recording NAV before and after. This is the central containment property: the Trading System can *trade* the portfolio but cannot *remove* it, since no code path lets its key transfer assets to an address of its choosing. The allowlist bounds the destination of value, not its amount, so a compromised Trading System could still trade the book badly through an approved router. It is a containment boundary, not a complete defence.

The share token is a standard ERC-20, so a fund position is transferable and composable. The vault tracks each holder's cost basis and a transfer hook carries that basis with the shares, so the performance fee follows the economics rather than the calendar: it is crystallised per investor at redemption, against their own entry price, on realised gains only.

The oracle stores a USDC-denominated price per token, writable only by a feeder role, and enforces no staleness requirement: every money-moving path pushes fresh prices in the same operation, so a stale price cannot reach a value-moving transaction. Trust already rests on an honest feeder, so a staleness check would add liveness failure modes without removing that assumption.

Authority is split across four roles: admin (role assignment, held by a multisig), manager (tracked tokens, allowlisted targets, fees, limits), bot (settlement and re-balancing only, the Trading System's hot wallet), and guardian (pause and emergency control). The role that runs continuously and is most exposed is the least privileged; the roles that can change parameters are used rarely and kept cold.

7 Why Web3

A quantitative manager and their investor want two things that appear to conflict. The manager must keep the strategy secret, because its parameters, universe, and entry logic are the entire product, and published they are arbitrated or copied. Full transparency is not a virtue here but a way of destroying the thing being sold. The investor must be able to verify they are not being defrauded: not that the strategy is good, which is their risk to take, but that reported NAV corresponds to real assets, that fees are as promised, that redemptions will be

honoured, and that the manager cannot abscond. Historically that verification is bought indirectly, at a cost and lag that only makes sense above a certain fund size.

These are claims about different things, the strategy's inputs and logic versus its outputs and custody, and a public ledger keeps the first private while making the second checkable by anyone, without asking us. Token balances at the vault address are public, so the assets are known to exist without a custodian's statement. NAV is computed by contract code from those balances, so the investor does not receive a valuation, they recompute it. Every rebalance is a permanent timestamped transaction, making selective disclosure of good months structurally impossible. Shares are minted and burned at a price the contract computes, not one the manager strikes. Fees are arithmetic in a published contract against a tracked cost basis, not an invoice. And a settled withdrawal is a claim against the contract, requiring no cooperation from us.

None of this reveals the strategy. An observer sees positions after the fact, roughly what a quarterly holdings disclosure gives in traditional markets, but not the signal that produced them, the universe considered, the forecasts consulted, or the conditions under which a position exits.

Verifiability answers "can I check what happened?". Containment answers "what is the worst that can happen?", and is the stronger property. The Trading System holds a necessarily online key, which must be assumed compromisable. Its role authorises trading through approved venues and settling the queue, and nothing else: it cannot transfer assets out, change the fee, add a target, alter the oracle, or grant itself a role. An attacker who takes the Trading System's key inherits the ability to trade badly, not to steal. The privilege gradient runs opposite to the exposure gradient, which is the property that matters.

HPT therefore does not claim to be trustless. It claims something narrower: that what an investor must take on faith has been reduced to the strategy's performance and a short list of operational assumptions, chiefly the honesty of the oracle feeder, the correctness of the contracts, and the custody of the governance keys, and that custody, valuation, issuance, fees, and redemption have been moved out of that set and into code anyone can read and execute themselves. That is the purpose of building on Web3: not decentralisation as an end, but the removal of intermediaries whose only function was to assure the investor of facts a public ledger now demonstrates directly.

8 Risks

Strategy and evidence. Trend-following has a low hit rate, a few large winners funding many small losses, and extended drawdowns in range-bound markets. Figure 1 shows the shape plainly: both panels are flat for most of 2023 and 2024, and most of the result arrives in a handful of moves, so an investor should expect long unprofitable stretches as normal operation. Table 1 also covers a single three-year window in one broadly favourable regime, and the EMA spans were not chosen blind to it. The result is encouraging but not yet out-of-sample.

Model. FinCast is pretrained on data that is not our universe, and its uncertainty estimates may be poorly calibrated in exactly the conditions where they matter most. The architecture limits the damage, since the model picks between two classical signals rather than generating positions, but a persistently miscalibrated σ would leave the strategy on the wrong lookback for extended periods.

Execution and liquidity. Depth that supported entry may not support exit, and tends to disappear precisely when an exit signal fires. On-chain execution also faces adversarial ordering, since pending transactions are visible and can be front-run. Given the cost sensitivity in Figure 1, this is the risk with the most direct effect on returns.

Technical and counterparty. A defect in any of the three contracts could cause loss regardless of strategy performance, and prices depend on a trusted feeder (\$7). If the Trading System stops, the vault holds assets safely but the queue stops being serviced. Holdings include wrapped and tokenised assets whose value depends on an issuer, and USDC is itself an issuer liability: exposures to entities, not markets.

9 Team and contact

Harris Partners Technologies is a one-person operation. Campbell Harris designed and built the Trading System, the Smart Contract Logic and the investor site, and operates the deployment. There is no other team, advisory board or outside investor; the “we” of this document is one person, and the reader should weigh key-person risk accordingly (\$8).

Campbell Harris, founder · casper@harrispartners.tech
linkedin.com/in/campbellharris2025 · pm.harrispartners.tech